

QUALITY CONTROL — OUT-OF-SPECIFICATION REPORT

Dissolution Testing | Batch B-101 | CAPA-VAL-003

Report No.:	QC-OOS-2026-0301	Report Date:	01-Mar-2026
Batch No.:	B-101	Product:	Tablet Formulation X
Test Method:	USP <711> Apparatus 2	Analyst:	J. Mehra, QC Lab
CAPA Ref.:	CAPA-VAL-003	Status:	FAILED — OOS CONFIRMED

1. Test Objective

This report documents the out-of-specification (OOS) dissolution result for Batch B-101 identified during routine stability testing. The test was conducted per USP <711> Apparatus 2 (paddle method) at 50 rpm in 900 mL phosphate buffer pH 6.8, 37 ± 0.5 °C. Specification requires ≥ 80% dissolved at 30 minutes.

2. Dissolution Results — 30-Minute Time Point

Vessel	% Dissolved (30 min)	Specification	Pass / Fail
V1	61%	≥ 80%	FAIL
V2	55%	≥ 80%	FAIL
V3	59%	≥ 80%	FAIL
V4	57%	≥ 80%	FAIL
V5	60%	≥ 80%	FAIL
V6	56%	≥ 80%	FAIL
Batch Average	58%	≥ 80%	OOS CONFIRMED

All six vessels failed the 30-minute dissolution specification. Batch average of 58% is 22 percentage points below the minimum acceptance criterion of 80%. The low and consistent failure across all vessels indicates a systematic batch-level issue rather than a single outlier.

3. Analytical Observations

Parameter	Observed Value	Specification / Target	Remark
Dissolution @ 30 min (avg)	58%	≥ 80%	OOS — 22% below spec
Tablet Hardness (avg)	4.1 kP	6.0 – 9.0 kP	Below minimum hardness target
Compression Force (avg)	4.5 kN	5.0 kN target	Setpoint drift documented
Tablet Weight Variation	Within spec	±5% of mean	Acceptable
Disintegration Time	Not tested	—	To be evaluated in Phase 2

4. Preliminary Root Cause Hypothesis

Based on cross-referencing dissolution data with manufacturing records, the following preliminary root causes are identified for CAPA investigation:

Priority	Cause ID	Description	Process Step	Evidence
1 (Primary)	C001	Compression force setpoint drift (4.3–4.5 kN vs. 5.0 kN target)	Compression	Logistics: 2 impacted tablets with low hardness
2 (Contributing)	C002	Granulation moisture excursion (+2.3% RH spike)	Granulation	Key: Granulation binder dissolution and reduced consolidation
3 (Possible)	C003	Broad API particle size distribution (D50 = 120 µm)	Milling	Supplier A: coarse tail reduction

5. Batch Disposition

Batch B-101 is placed on HOLD pending completion of CAPA investigation CAPA-VAL-003. No further distribution, release, or use of this batch is permitted. All retained samples have been quarantined under controlled storage conditions. Re-test of reserve samples is authorized upon QA approval only.

6. Linked CAPA & Historical Context

Historical CAPA-2024-018 previously identified low compression force as a dissolution failure mechanism in pilot batches. Corrective actions included re-calibration of the compression press and operator retraining. Investigation record inv.1 confirms that the pre-run calibration check (SOP-COMP-004) was skipped on the day of manufacture due to staffing shortages, representing a recurrence of the root cause addressed in CAPA-2024-018.

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